LLMLingua: Compressing Prompts for Accelerated Inference of Large Language Models

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EMNLP 2023

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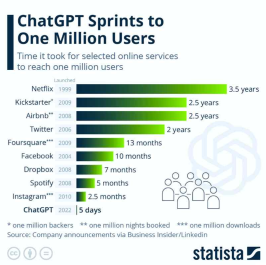
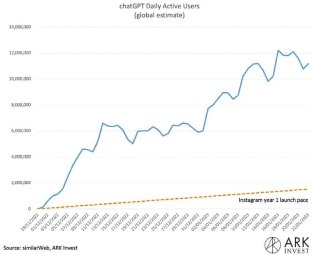
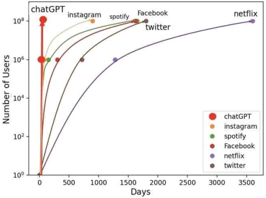
2023.10.29

ChatGPT's infleunce



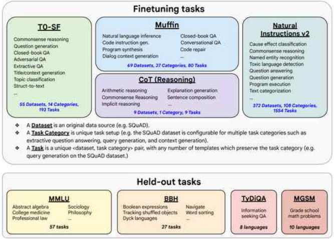


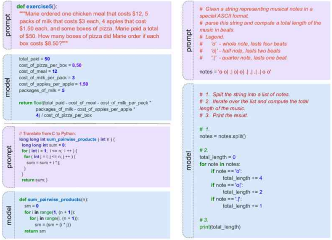
ChatGPT's influence



ChatGPT’s infleunce

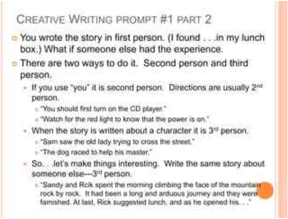
- Powerful generalization and reasoning capabilities of LLMs.
- Suitable prompts is crucial, they involves techniques such as chain-of-thought, in-context learning, and retrieving related documents ([Wei et al., 2022](); [Chase, 2022]()).

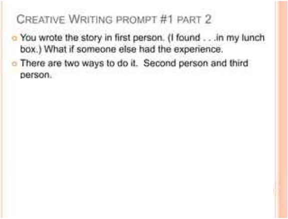




Background

- However, activating LLM’s domian specific knowledge, they require long prompts (i.e. in-context learning, chain-of-thought).
- Approaches that attempt to reduce the length of original prompts have emerged lately while preseving essential information.





Background

- Removing tokens with lower perplexity has a relatively minor impact on the LLM's comprehesion of the context.
- Motivated by this, Li(2023) propose Selective-Context.
  - compute self-information of lexical unit (sentences, phrases, or tokens) and drop the less informative content

Context: Large Languages Models (LLMs) {~~*trained on massive corpora of texts*~~} have shown their ability to perform new tasks from textual instructions or from a few examples, which resulting in a line of work that focuses on further scaling these models. These efforts are based on the assumption {~~*that more parameters will lead to better performance.*~~}

Query: What's the assumption behind?

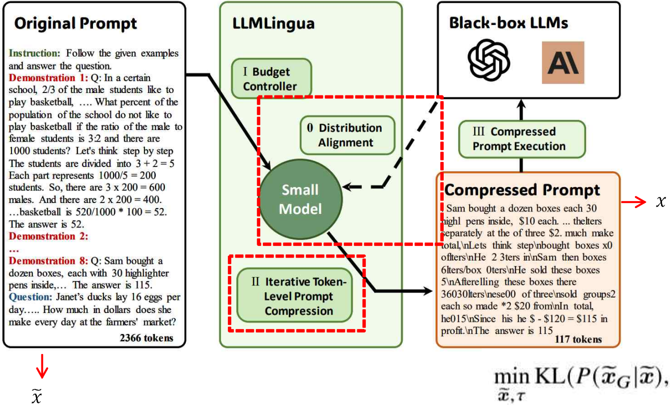
GPT: Further scaling Large Language Models will lead to better performance on a wide range of tasks.

self-information

$$I(x_i) = -\log_2 P(x_i | x_0, x_1, ..., x_{i-1})$$

Limit : ignore the independence between correspondense contents
& neglect correspondence between the LLM and the small language model

Problem Fomulation



$$\min_{\widetilde{\boldsymbol{x}},\tau} \mathrm{KL}(P(\widetilde{\boldsymbol{x}}_G|\widetilde{\boldsymbol{x}}), P(\boldsymbol{x}_G|\boldsymbol{x})),$$

pre-defined tau

Primary knowledge

- PPL (Perplexity)
  - One of the most common metric for evaluating language models.
  - Perplexity is defined as the <u>exponentiated average negative log-likelihood</u> of a sequence.
  - In terms of information entropy, tokens with lower PPL contribute less to the entropy gains of the language model.

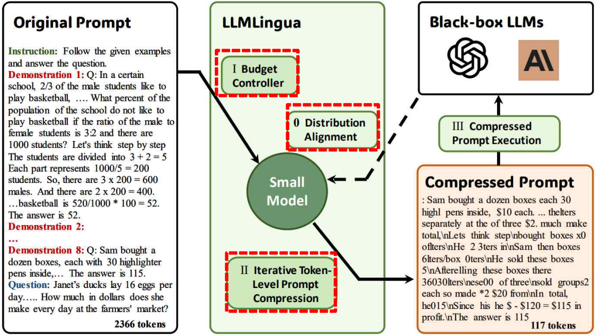
Hugging Face is a startup based in New York City and Paris

p(word|context)

$$\mathrm{PPL}(\mathrm{X}) = exp\left\{-\frac{1}{t}\sum_{i}^{t} log p_{\theta}(x_i|x_{<i})\right\}$$

X : a tokenized sequence

LLMLingua - 3 steps



0. Distribution Alignment

- To narrow the gap between the distribution LLM and that of small LM.
- Aligning the two distribution <u>via instruction tuning</u>
- Pre-trained small language model M_s
- A pair of instruction x_i and LLM generated text $y_{i,LLM}$

$$\min_{\boldsymbol{\theta}_s} \mathbb{E}\left[\frac{1}{N}\sum_{i=1}^{N}\mathcal{L}\left(\mathbf{x}_i, \mathbf{y}_{i,\mathrm{LLM}}; \boldsymbol{\theta}_{\mathcal{M}_s}\right)\right]$$

I. Buget Controller

- Allocate different bugets
  (i.e. instructions, demonstrations, and questions)

- Two considerations

  - Instruction and question - direct influence on the generated results , less rebundant.
    > Smaller compression ratios
  - High compression ratio - sentence-level dropout with preserving a linquistic integrity.
    > Multiple demonstration : demonstration-level control

I. Buget Controller

Algorithm 1 Pseudo code of Budget Controller.

Input: A small language model $\mathcal{M}_s$; the original prompt $\boldsymbol{x} = (\boldsymbol{x}^{\text{ins}}, \boldsymbol{x}^{\text{dems}}, \boldsymbol{x}^{\text{que}})$.

Set the selected demonstration set $\mathcal{D} = \phi$.
Get demonstration compression rate τ_{dem} by Eq.(2).
Calculate the perplexity of each demonstration via $\mathcal{M}_s$.
Rank all demonstrations in descending order of their perplexity as a list $(\boldsymbol{x}^{\text{dem}}_{(1)}, ..., \boldsymbol{x}^{\text{dem}}_{(N)})$, where N is the number of demonstrations, $\boldsymbol{x}^{\text{dem}}_{(i)}$ is the i-th demonstration.
for $i = 1$ **do**
  if $\widetilde{L}_{\mathcal{D}} > k \cdot \tau_{\text{dems}} L_{\text{dems}}$ **then**
    Break.
  end if
  Append $\boldsymbol{x}^{\text{dem}}_{(i)}$ to $\mathcal{D}$.
  $i = i + 1$
end for
Allocate remaining budget to $\boldsymbol{x}^{\text{ins}}$ and $\boldsymbol{x}^{\text{que}}$ via Eq.(3).

Output: The subset of demonstrations $\mathcal{D}$ obtained from coarse-grained compression; Additional budget $\Delta\tau_{\text{ins,que}}$ for the instruction and the question.

total length

pre-defined tau

$$\tau_{\text{dems}} = \frac{\tau L - (\tau_{\text{ins}} L_{\text{ins}} + \tau_{\text{que}} L_{\text{que}})}{L_{\text{dems}}}.$$

$$0.696 = \frac{0.7 \times 1855 - (0.85 \times 30 + 0.9 \times 25)}{1800}$$

control coefficient

$$\Delta\tau = \frac{k \cdot \tau_{\text{dems}} L_{\text{dems}} - \widetilde{L}_{\mathcal{D}}}{L_{\text{ins}} + L_{\text{que}}},$$

coressponding compression ratio

$$= \frac{k \cdot 1252.8 - 1260}{55}$$

II. Iterative Token-level Prompt Compression

- Perplexity encounters the limitation i.e., the independence assumption.
- Original prompt after demonstration-level compression : x’ $\longrightarrow$ $\mathrm{S} = \{\mathrm{s}_1, \mathrm{s}_2, \ldots, \mathrm{s}_m\}$
- Each segment is concatenated to the subsequent segment, conditional probabilites for each segment $\mathrm{p}(\widetilde{s}_j)$ are obatained.

$$p(\widetilde{\boldsymbol{x}}) = \prod_{i=1}^{\widetilde{L}} p(\widetilde{x}_i|\widetilde{x}_{<i})$$

$$\approx p(\boldsymbol{x}') = \prod_{i=1}^{L'} p(\underline{x_i|\widetilde{x}_{<i}, \overline{x}_{<i}}),$$

$$p(\widetilde{\boldsymbol{s}}_j) = \prod_{i=1}^{\sum_k^j \widetilde{L}_{s,k}} p(\widetilde{s}_{j,i}|\widetilde{s}_{j,<i}, \widetilde{\boldsymbol{s}}_{<j})$$

$$\approx \prod_{i=1}^{L_{s,j}+\sum_k^{j-1} \widetilde{L}_{s,k}} \boxed{p(s_{j,i}|s_{j,<i}, \widetilde{\boldsymbol{s}}_{<j}),}$$

II. Iterative Token-level Prompt Compression

- Perplexity is based on the independence assumption.
- Original prompt after demonstration-level compression : x' $\longrightarrow$ $S = \{s_1, s_2, \ldots, s_m\}$
- Each segment is concatenated to the subsequent segment, conditional probabilites for each segment $p(\tilde{s}_j)$ are obatained.
- The compression ratio threshold γ_j are dynamically caculated based on the PPL
- tokens in each s_j with the PPL greater than γ_j are retained in compressed prompt.

$$\tau_{s_j} = \begin{cases} \tau_{\text{ins}} + \Delta\tau, & \text{if } s_j \text{ from } x^{\text{ins}}, \\ \tau_{\text{dems}}, & \text{if } s_j \text{ from } x^{\mathcal{D}}, \\ \tau_{\text{que}} + \Delta\tau, & \text{if } s_j \text{ from } x^{\text{que}}. \end{cases}$$

coressponding compression ratio

$$\widetilde{s}_j = \{s_{j,i} \boxed{p(s_{j,i}) > \gamma_j}\}$$

II. Iterative Token-level Prompt Compression

Algorithm 2 Pseudo code of Iterative Token-level Prompt Compression (ITPC).

Input: A small language model $\mathcal{M}_s$; the prompt from budget controller $\boldsymbol{x}' = (\boldsymbol{x}^{\text{ins}}, \boldsymbol{x}^{\mathcal{D}}, \boldsymbol{x}^{\text{que}})$; target compression rate τ, adjusted compression rate $\Delta\tau_{\text{ins,que}}$.

```
Set the selected token set 𝒯 = ϕ
Get segment set 𝒮.
for i = 1, 2, ..., m do
    Get the conditional probabilities p(s_i) via Eq.(5)
    Get the compression threshold γ_i with Eq. (6)
    Append the compressed token to 𝒯 via Eq.(7)
end for
Concatenate all tokens in 𝒯 as x̃.
```

Output: The compressed prompt $\widetilde{\boldsymbol{x}}$.

$$p(\widetilde{\boldsymbol{s}}_j) = \prod_{i=1}^{\sum_k^j \widetilde{L}_{s,k}} p(\widetilde{s}_{j,i} | \widetilde{s}_{j,<i}, \widetilde{\boldsymbol{s}}_{<j})$$

$$\approx \prod_{i=1}^{L_{s,j}+\sum_k^{j-1} \widetilde{L}_{s,k}} p(s_{j,i} | s_{j,<i}, \widetilde{\boldsymbol{s}}_{<j}),$$

$$\tau_{\boldsymbol{s}_j} = \begin{cases} \tau_{\text{ins}} + \Delta\tau, & \text{if } \boldsymbol{s}_j \text{ from } \boldsymbol{x}^{\text{ins}}, \\ \tau_{\text{dems}}, & \text{if } \boldsymbol{s}_j \text{ from } \boldsymbol{x}^{\mathcal{D}}, \\ \tau_{\text{que}} + \Delta\tau, & \text{if } \boldsymbol{s}_j \text{ from } \boldsymbol{x}^{\text{que}}. \end{cases}$$

$$\widetilde{\boldsymbol{s}}_j = \{s_{j,i} | p(s_{j,i}) > \gamma_j\}$$

Q & A

Experimental Settings

- **Data**

ShareGPT	ShareGPT is a dataset which users share conversations with ChatGPT indifferent languages and invarious scenario (i.e. chitchat, assistant and code)	
	yn2eWCt	[["human", "how do I add multiple new columns in m for power query or power bi?"], ["gpt", "In Power Query or Power BI, you can add multiple new columns by using the \"Add Column\" menu in the Home tab. Once you click on the \"Add Column\" menu, you will ...
Arxiv March 2023	A dataset consisting of latest academic papers created in March 2023 from the arXiv preprint repository.	

http://arxiv.org/abs/2303.06806v1	20230313012855	Neural Diarization with Non-autoregressive Intermediate Attractors	["Yusuke Fujita", "Tatsuya Komatsu", "Robin Scheibler", "Yusuke Kida", "Tetsuji Ogawa"]	eess.AS	["eess.AS", "cs.CL", "cs.SD"]	Boosting Source Code Learning with Data Augmentation: An Empirical Study Jianjun Zhao ====== End-to-end neural diarization (EEND) with encoder-decoder-based attractors (EDA) is a promising method to handle the whole speaker diarization problem simultaneously with a single neural network. While the EEND model can produce all fram ...

Experimental Settings

- **Data**

GSM8K

GSM8K is a dataset of 8.5K high quality linguistically diverse grade school math word problems created by human problem writers. The dataset is segmented into 7.5K training problems and 1K test problems.

Natalia sold clips to 48 of her friends in April, and then she sold half as many clips in May. How many clips did Natalia sell altogether in April and May?	Natalia sold 48/2 = <<48/2=24>>24 clips in May. Natalia sold 48+24 = <<48+24=72>>72 clips altogether in April and May. #### 72

BBH

BIG Benchmark (**B**eyond the Imitation Game **B**enchmark) In traditional NLP, 200 diverse text-based tasks such as mathematics, commonsense reasoning, and question-answering. BIG-Bench Hard (BBH) is a subset of 23 particularly **challenging BIG-Bench tasks**.
- Multiple-choice qa, open-domain qa, multi-label classification etc.

Input	Target
True and not not (not False) is	True

Experimental Settings

- **Baselines**

- **GPT-4 Generation**
  - instruct the GPT-4 to compress the original prompt.
- **Random Selection**
  - randomly selecting the demonstrations or sentences of the original prompt
- **Selective-Context (Li, 2023)**
  - phraprase-level self information to filter out less informative content.
    - same small LM (i.e. Alpaca-7B)

Main Results

compression ratio

Methods	ShareGPT							Arxiv-March23						
	BLEU	Rouge1	Rouge2	RougeL	BS F1	Tokens	$1/\tau$	BLEU	Rouge1	Rouge2	RougeL	BS F1	Tokens	$1/\tau$
Constraint I	*2x constraint*							*350 tokens constraint*						
Sentence Selection	**28.59**	46.11	**31.07**	37.94	88.64	388	1.5x	22.77	50.1	25.93	33.63	88.21	379	4x
Selective-Context	25.42	46.47	29.09	36.99	88.92	307	1.9x	21.41	51.3	27.94	36.73	89.60	356	4x
Ours	27.36	**48.87**	30.32	**38.55**	**89.52**	304	1.9x	**23.15**	**54.21**	**32.66**	**42.74**	**90.33**	345	4x
Constraint II	*3x constraint*							*175 tokens constraint*						
Sentence Selection	18.94	35.17	18.96	26.75	85.63	255	2.3x	12.41	38.91	14.25	26.72	87.09	229	7x
Selective-Context	15.79	38.42	20.55	28.89	87.12	180	3.3x	12.23	42.47	19.48	29.47	88.16	185	8x
Ours	**19.55**	**40.81**	**22.68**	**30.98**	**87.70**	177	3.3x	**13.45**	**44.36**	**24.86**	**34.94**	**89.03**	176	9x
		(+2.41)	(+2.13)							(+5.38)	(+5.47)			9x

Main Results

Methods	GSM8K			BBH		
	EM	Tokens	$1/\tau$	EM	Tokens	$1/\tau$
Full-shot	78.85	2,366	-	70.07	774	-
1-shot constraint						
1-shot	77.10	422	6x	69.60	284	3x
Selective-Context	53.98	452	5x	54.27	276	3x
GPT4 Generation	71.87	496	5x	27.13	260	3x
Ours	**79.08**	446	5x	**70.11**	288	3x
half-shot constraint						
Sentence Selection	72.33	230	10x	39.56	175	4x
Selective-Context	52.99	218	11x	54.02	155	5x
GPT4 Generation	68.61	223	11x	27.09	161	5x
(-1.44) **Ours**	**77.41**	171	14x	**61.60**	171	5x
quarter-shot constraint						
Sentence Selection	66.67	195	12x	46.00	109	7x
Selective-Context	44.20	157	15x	47.37	108	7x
GPT4 Generation	56.33	188	20x	26.81	101	8x
(-1.52) **Ours**	**77.33**	117	20x	**56.85**	110	7x
zero-shot	48.75†	11	215x	32.32	16	48x
Simple Prompt	74.9	691	3x	-	-	-

Main Results

Methods	GSM8K EM	GSM8K Tokens	GSM8K $1/\tau$	BBH EM	BBH Tokens	BBH $1/\tau$
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Simple Prompt	74.9	691	3x	-	-	-

BBH

challenging resoning tasks -> drop

Analysis

Compressed Prompt:
Follow the given examples and answer the question.
: Sam bought a dozen boxes each 30 highl pens inside, $10 each. He reanged five of boxes into of six each $3 per. He sold the thelters separately at the of three $2. much make total,
Lets think step
bought boxes x0 oflters
He 2 3ters in
Sam then boxes 6lters/box 0ters
He sold these boxes 5
Afterelling these boxes there 36030lters
ese00 of three
sold groups2 each so made *2 $20 from
In total, he015
Since his he $ - $120 = $115 in profit.
The answer is 115

Q: Toula went to the bakery and bought various types of pastries. She bought 3 dozen donuts which cost $68 per dozen, 2 dozen mini cupcakes which cost $80 per dozen, and 6 dozen mini cheesecakes for $55 per dozen. How much was the total cost?

Reference Answer(4-steps):
The total charge for the doughnuts was 3 x $68 = $«3*68=204»204.
The total charge for the mini cupcakes was 2 x $80 = $«2*80=160»160.
The total charge for the mini cheesecakes was 6 x $55 = $«6*55=330»330.
Therefore the total amount Toula paid for the pastries was $204 + $160 + $330 = $«204+160+330=694» 694.

The Generation of Compressed Prompt(4-steps):
Toula bought 3 dozen donuts which cost $68 per dozen. So, she spent 3 x 68 = $204 on donuts.
She also bought 2 dozen mini cupcakes which cost $80 per dozen. So, she spent 2 x 80 = $160 on mini cupcakes.
Finally, she bought 6 dozen mini cheesecakes for $55 per dozen. So, she spent 6 x 55 = $330 on mini cheesecakes.
Therefore, the total cost was 204 + 160 + 330 = $694.
The answer is 694.

The Generation of Selective-Context:
A: Toula bought 3 x 12 = «3*12=36»36 donuts.
Then, the total cost of the donuts is 36 x $68 = $«36*68=2448»2448.
How many mini cupcakes did Toula buy in total?
She bought 2 x 12 = «2*12=24»24 mini cupcakes.
So, the total cost of the mini cupcakes is 24 x $80 = $«24*80=1920»1920.
How many mini cheesecakes did Toula buy in total?
She bought 6 x 12 = «6*12=72»72 mini cheesecakes.
Thus, the total cost of the mini cheesecakes is 72 x $55 = $«72*55=3960»3960.
Therefore, the total cost of all the pastries is $2448 + $1920 + $3960 = $«2448+1920+3960=8328»8328. Answer: 8328.

- Especially on GSM8K, its performance is lower than ours by 33.10 points at a compression ratio of 20x.

Ablation

	EM	Tokens	$1/\tau$
Ours	**79.08**	439	5x
- w/o Iterative Token-level Prompt Compression	72.93	453	5x
- w/o Budget Controller	73.62	486	5x
- w/o Dynamic Compression Ratio	77.26	457	5x
- w/ Random Selection in Budget Controller	72.78	477	5x
- w/o Distribution Alignment	78.62	452	5x
- w/ Remove Stop Words	76.27	1,882	1.3x

Table 3: Ablation study on GSM8K in 1-shot constraint.

- They conjecture this variant may lose essential information in the prompt, especially for lowfrequency keywords that frequently appear in thegiven prompt.
- Theirs w/ Random Selection in Budget Controller indicates that selecting sentences or demonstrations based on perplexity can better identify information-rich sentences for target LLMs.

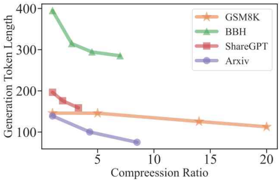
Discussion

- **Three questions :**
  - Different Target LLM > **Claude-v1.3**
  - Different Small LMs > **GPT-2, Alpaca-7B**
  - Generation result of compressed prompt

    > **compression makes savings in generation stage!**

	EM	Tokens	$1/\tau$
Ours in 1-shot constraint	83.51	439	5x
Ours in half-shot constraint	82.61	171	14x
Simple Prompt	81.8	691	3x

	EM	Tokens	$1/\tau$
Ours with GPT2 in 1-shot constraint	77.02	447	5x
Ours with GPT2 in half-shot constraint	76.42	173	14x
Ours with GPT2 in quarter-shot constraint	76.27	128	18x



They insist the generalization of method.

Small model for compression show performance on Alpaca-7B than GPT-small!

Conclusion

- They introduce LLMLingua, which is based on the small LM's PPL for black-box LLMs.
- They validate the effectiveness of their approach on 4 datasets from different domains, i.e., GSM8K, BBH, ShareGPT, and Arxiv-March23.
- Thier approach holds substantial practical implications, as it not only reduces computational costs but also offers a potential solution for accommodating longer contexts in LLMs.

Limitation

- They introduce LLMLingua, which is based on the small LM's PPL for black-box LLMs.
- They validate the effectiveness of their approach on 4 datasets from different domains, i.e., GSM8K, BBH, ShareGPT, and Arxiv-March23.
- Thier approach holds substantial practical implications, as it not only reduces computational costs but also offers a potential solution for accommodating longer contexts in LLMs.

Demos

Instruction

Please reference the following examples to answer the math question,

Context

Question: Angelo and Melanie want to plan how many hours over the next week they should study together for their test next week. They have 2 chapters of their textbook to study and 4 worksheets to memorize. They figure out that they should dedicate 3 hours to each chapter of their textbook and 1.5 hours for each worksheet. If they plan to study no more than 4 hours each day, how many days should they plan to study total over the next week if they take a 10-minute break every hour, include 3 10-minute snack breaks each day, and 30 minutes for lunch each day?
Let's think step by step
Angelo and Melanie think they should dedicate 3 hours to each of the 2 chapters, 3 hours x 2 chapters = 6 hours total.
For the worksheets they plan to dedicate 1.5 hours for each worksheet, 1.5 hours x 4 worksheets = 6 hours total.
Angelo and Melanie need to start with planning 12 hours to study, at 4 hours a day, 12 / 4 = 3 days.
However, they need to include time for breaks and lunch. Every hour they want to include a 10-minute break, so 12 total hours x 10 minutes = 120 extra minutes for breaks.
They also want to include 3 10-minute snack breaks, 3 x 10 minutes = 30 minutes.
And they want to include 30 minutes for lunch each day, so 120 minutes for breaks + 30 minutes for snack breaks + 30 minutes for lunch = 180 minutes, or 180 / 60 minutes per hour = 3 extra hours.
So Angelo and Melanie want to plan 12 hours to study + 3 hours of breaks = 15 hours total.
They want to study no more than 4 hours each day, 15 hours / 4 hours each day = 3.75
They will need to plan to study 4 days to allow for all the time they need.
The answer is 4

Question: Mark's basketball team scores 25 2 pointers, 8 3 pointers and 10 free throws. Their opponents score double the 2 pointers but half the 3 pointers and free throws. What's the total number of points scored by both teams added together?
Let's think step by step
Mark's team scores 25 2 pointers, meaning they scored 25*2= 50 points in 2 pointers.

Question

Question: Josh decides to try flipping a house. He buys a house for $80,000 and then puts in $50,000 in repairs. This increased the value of the house by 150%. How much profit did he make?

Target Token (To use this, set Compression Ratio to 0)

200

Compression Ratio (To use this, set Target Token to -1)

0

Compress Prompt!

Demos

Compressed Prompts

compressed_prompt

Please reference the following examples to answer the math question,

Question: can buy 4 apples 1 for You bought 36 fruits evenly split between and of 1 $ does cost if total bill $
's think step
If were between 3 of then I 363 = 12 of fruitIf 1 orange then oranges50 $If I I $66 $60 the 2 fruit
of is, and is then 1W=4If we know we bought 12 and 12W 12
that can the 12 =
of apple (60/The
: Sam a dozen with 30 highighter pens each Heanged into3 the separately the. much
think boxes120 12 = 360Sam then took 5 boxes × 6 highlighters/box = 30 highlighters.
He sold these boxes for 5 * $3 = $15
After selling these 5 boxes there were 360 - 30 = 330 highlighters remaining.
These form 330 / 3 = 110 groups of three pens.
He sold each of these groups for $2 each, so made 110 * 2 = $220 from them.
In total, then, he earned $220 + $15 = $235.
Since his original cost was $120, he earned $235 - $120 = $115 in profit.
The answer is 115

Question: Josh decides to try flipping a house. He buys a house for $80,000 and then puts in $50,000 in repairs. This increased the value of the house by 150%. How much profit did he make?

Saving

The tokens number of original prompt

2428

The tokens number of compressed prompt

331

Actual Compression Ratio

7.3x

Saving Cost

, Saving $0.1 in GPT-4.

Open Source

```
instruction ="""- Title: Writing questions that involve commonsense understanding of event duration.
- Definition: In this task, we ask you to write a question that involves ?event
duration", based on a given sentence. Here, event duration is defined as the
understanding of how long events typically last. For example, ?brushing teeth?, usually takes few minutes.
- Emphasis & Caution: The written questions are not required to have a single
correct answer.
- Things to avoid: Don't create questions which have explicit mentions of
answers in text. Instead, it has to be implied from what is given. In other words, we want you to use "instin
"""

context = """Positive Example -Input: Sentence: Jack played basketball after school, after which he was very
Negatvie Example -Input: Sentence: He spent two hours on his homework. -Output: How long did he do his homewo
Negatvie Example -Input: Sentence: It's hail crackled across the comm, and Tara spun to retake her seat at th
Negatvie Example -Input: Sentence: There was even a tiny room in the back of one of the closets. ssssss-Expec

question = """Ask a question on "event duration" based on the provided sentence."""
```

Demonstration original length : [53, 51, 42, 49]

[[(9, 0.15306122448979592)], [(10, 0.15306122448979592)], [(10, 0.15306122448979592)], [(10, 0.15306122448979592)], [(10, 0.15306122448979592)], [(4, 0.15306122448979592), (6, 0.15306122448979592)], [(10, 0.15306122448979592)], [(10, 0.15306122448979592)], [(10, 0.15306122448979592)], [(10, 0.15306122448979592)], [(5, 0.15306122448979592), (5, 0.15306122448979592)], [(10, 0.15306122448979592)], [(10, 0.15306122448979592)], [(10, 0.15306122448979592)], [(7, 0.15306122448979592), (3, 0.15306122448979592)], [(10, 0.15306122448979592)], [(10, 0.15306122448979592)], [(10, 0.15306122448979592)], [(10, 0.15306122448979592)], [(6, 0.15306122448979592)]]

After Budget Controller - iterative size : “10”

```
Threshold
25.647850036621094
Compressed input ids
[": : played basketball after school, after which he was very tired. - output : how long did jack play basketball? - reason : the question asks about the duration of an event ; therefore it's a temporal event duration question. negatvie example - input
 : sentence : he spent two hours on his homework. - output : how long did he do his homework? - reason : we donot want this question as the answer is directly mentioned in the text. - suggestion : - negatvie example - input : sentence : it's hail crack
led across the comm, and tara spun to retake her seat at the helm. - expected output : how long was the storm? negatvie example - input : sentence : there was even a tiny room in the back of one of the closets. ssssss - expected output : after buying t
he house, how long did it take the owners to notice the room? [SEP]"]
```

1 step on ITPC

“Drop ‘positive example, jack’”

Q & A

Thank You!